

Supporting Information

Inference-Sufficient Representations for High-Throughput Measurement: Lessons from Lossless Compression Benchmarks in 4D-STEM

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This document records the parameters of the simulated 4D-STEM dataset shown in Figure 6 of the main text and the calculations behind Table 6. The simulated data are illustrative; no result of the compression benchmark depends on them.

S1 The simulated dataset

S1.1 Specimen

A pristine monolayer of 2H-WSe₂, built with `ase.build.surface` from the (001) surface, one layer thick, and converted to an orthogonal supercell by the transformation $[[-2, -1, 0], [0, -1, 0], [0, 0, 1]]$ with 3×6 lateral repeats. The result is 108 atoms—36 tungsten and 72 selenium—in a $17.043 \times 19.680 \times 7.340$ Å cell, with 2 Å of vacuum on each side of the slab. The in-plane lattice parameter is 3.28 Å and the Se–Se separation across the layer is 3.34 Å.

There are no defects, no substitutions and no adjusted scattering weights. Species-specific Kirkland parameterizations are used for both elements.

S1.2 Multislice calculation

Thermal displacement is applied as a frozen-phonon average over intensity rather than amplitude. The quoted σ of Table S1 is the root-mean-square displacement *magnitude*; each Cartesian component is drawn independently from $\mathcal{N}(0, \sigma/\sqrt{3})$. The configurations are generated from a fixed seed (20260916) and one thermal state is used for the whole scan, so that every probe position sees the same specimen.

S1.3 Scan and detector

The probe is rastered over 128×128 positions spanning the cell, a step of 0.1332×0.1537 Å. The probe-limited Nyquist step is 0.348 Å, so the scan is oversampled by $2.61\times$ and $2.26\times$ along the two axes and adjacent probes overlap by 92% along the fast axis.

Table S1: Multislice parameters.

Accelerating voltage	80 kV
Relativistic wavelength	0.041756 Å
Convergence semi-angle	30 mrad
Probe diameter ($1.22\lambda/\alpha$)	1.698 Å
Probe aberrations	none (ideal, in focus)
Anti-aliasing band limit	180 mrad
Real-space grid	221×255
Real-space sampling	0.07712×0.07718 Å
Slices	16
Slice thickness	0.45875 Å
Frozen-phonon configurations	16
Frozen-phonon displacement	$\sigma = 0.09$ Å

Each diffraction pattern is recorded on a 147×169 grid of 24,843 pixels spanning ± 178.9 and ± 178.2 mrad. The pixel is anisotropic, 2.450×2.122 mrad, because the orthogonal supercell is not square; the detector panels of Figure 6 are therefore plotted in mrad rather than on pixel indices, on which a circular aperture would render as an ellipse. The annular dark-field detector spans 75–130 mrad.

S1.4 Dose and counting

A beam current of 60 pA and a dwell of 1 μ s deliver a mean of 374.49 electrons to each probe position. This is a rate, not a count: the number arriving in any one dwell is itself a Poisson variable.

The simulation emits two quantities at every probe position. The first is the noiseless expected intensity λ , one real number per detector pixel, obtained by scaling the multislice intensity by the dose *divided by the incident total* rather than by each pattern’s own sum. Dividing by the pattern sum would assert that every electron lands somewhere on the detector, which is false: electrons scattered beyond the anti-aliasing band limit are not represented. Consequently $\sum \lambda$ is position-dependent and less than the dose—368.29 electrons at a tungsten column, where 1.7% is lost past the band limit, against 374.41 at a hollow site.

The second quantity is one Poisson draw from λ , taken independently at each pixel from a fixed seed (20260921). The frame total is therefore itself Poisson rather than fixed, and may exceed the dose: the patterns at a tungsten column (shown in Figure 6), a selenium column and a hollow site drew 347, 379 and 356 electrons against expectations of 368.3, 374.1 and 374.4. Over the full cube, 4,496,783 of 407,027,712 pixels are non-zero (1.10%) and the largest single count is 9.

S2 Code-length calculations

Two of Table 6’s five entries are direct products and are not derived below. The raw datacube is $16,384$ probe positions $\times$ $24,843$ detector pixels at one byte each, or 407,027,712 B (388.2 MiB); the ADF image is one float32 per position, 65,536 B (64.0 KiB). The three remaining entries follow.

S2.1 The shortest code given λ

Counts are independent across detector pixels given λ , so the entropy of the cube conditional on λ is the sum of the per-pixel Poisson entropies [1], with no approximation. This quantity is accumulated during the scan, one probe position at a time, so that the 1.6 GB of λ is never held in memory or written to disk.

The Poisson entropy has no elementary closed form, so it is summed directly rather than approximated. Each per-pixel entropy is evaluated by the recurrence $p_k = p_{k-1}\lambda/k$ from $p_0 = e^{-\lambda}$, truncated at $\lambda + 8\sqrt{\lambda} + 8$ terms. Evaluated against `scipy.stats.poisson` the agreement is better than 10^{-14} absolute over rates from 0 to 25.

Two numbers result. The expected code length, $\sum_i H(\text{Poisson}(\lambda_i))$, is 1,938,754 bytes. The length this particular Poisson draw actually costs a coder holding λ , $\sum_i -\log_2 P(c_i \mid \lambda_i)$, is 1,938,346 bytes. The two differ by 408 bytes, a relative difference of 2.11×10^{-4} , which is the expected scale of fluctuation over 4×10^8 pixels. Table 6 reports the second, being what a coder emits for this cube rather than its expectation over draws.

S2.2 The generating description

The 347 bytes quoted in the main text are the specimen and instrument parameters serialized as minified JSON, keys and decimal digits included, with no attempt at packing (Table S2). It is therefore an upper bound on the generating description, which is the conservative direction for the argument it supports.

Table S2: The generating description, as counted.

Specimen	compound, polytype, a , Se–Se separation, layers, supercell transformation, lateral repeats, vacuum	153 B
Instrument	voltage, convergence semi-angle, band limit, beam current, dwell, phonon σ and configuration count, real-space grid, slice count, scan shape	194 B
Total		347 B

Three classes of quantity are excluded. *Derived* values are omitted because the simulation computes them from the parameters above: the wavelength from the voltage, the probe diameter, the slice thickness from the cell height and slice count, the real-space sampling, the scan step, and the dose from the beam current and dwell. The *ADF inner and outer angles* are omitted because they define a virtual detector applied to the cube rather than anything needed to produce it. The *two random seeds* are omitted because they record which draw was taken rather than any fact about the specimen; adding them, sixteen further bytes, would make the cube reproducible bit-for-bit, which is a statement about the draw and not about the sample.

S2.3 Comparison with an off-the-shelf implementation

The Blosc Zstd entry of Table 6 uses the configuration benchmarked in the main text: the `hdf5plugin` Blosc filter with the zstd codec at compression level 3 and byte shuffling, on balanced chunks of (16, 16, 128, 128), with the cube stored as a $128 \times 128 \times 147 \times 169$ array. The figure quoted is the size of the resulting HDF5 file on disk, so it includes container and header overhead, and is directly comparable with the ratios reported for the measured datasets.

The model-based entry is not a file size. It is a code length, and carries no container, no header, and none of the small overhead a working arithmetic coder pays to flush its state.

S3 Software and reproduction

The simulation, the code-length routines and the script that produces Table 6 are all in the manuscript’s repository, together with the commands that run them and a README describing the sequence. Running the simulation takes a few minutes for a 128×128 scan on an NVIDIA RTX A5000, and reproduces the published cube bit-for-bit; the counts, the enlarged λ patterns, the virtual images and the atomic coordinates all match by SHA-256. A verification script in the repository performs that comparison against digests stored alongside it and exits non-zero on any difference.

Generating the dataset requires Python 3.12 and a CUDA device. Regenerating Table 6 from it, and verifying it against the stored digests, need NumPy alone.

The multislice physics is PySlice [2], an MIT-licensed code which, beyond the vibrational spectroscopy of its title, also performs conventional imaging and diffraction calculations. It is pinned at tag `4dstem-compression-2026` of <https://github.com/sea-ecosystem/PySlice>. That version and every other dependency are recorded in the repository’s lock file, so a reader obtains the same package versions this work used rather than whatever is current.

References

- [1] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, August 1991.
- [2] Harrison A. Walker, Thomas W. Pfeifer, Paul M. Zeiger, Jordan A. Hachtel, Sokrates T. Pantelides, and Eric R. Hoglund. PySlice: Routine Vibrational Electron Energy-Loss Spectroscopy Prediction with Universal Interatomic Potentials. *npj Computational Materials*, July 2026. doi:10.1038/s41524-026-02222-7.